
IM0802 – Responsible Artificial Intelligence

Oversight Without Control: A Comprehensive Human Oversight Analysis of the Habsora AI Targeting System

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Abstract

Habsora (‘The Gospel’) is an AI-targeting system deployed by the Israel Defense Forces (IDF) that generates over one hundred strike recommendations per day, each reviewed by a commander in approximately twenty seconds. Although each recommendation is formally approved by a human operator, this compression of human judgement into a near-instantaneous approval means genuine oversight is largely absent. This paper applies Verdiesen et al. [1]’s Comprehensive Human Oversight Framework (CHOF) to analyse Habsora across its technical, socio-technical, and governance layers. The CHOF treats human oversight not as a single requirement but as a structured set of conditions that can be satisfied or violated independently at each layer. Our analysis finds that Habsora fails across all three. We argue this is a structural design failure rather than operator intent, with consequences cascading across layers. At the technical layer, ensemble opacity prevents operators from understanding the basis of any recommendation. At the socio-technical layer, time pressure and absent explainability produce automation bias, reducing formal human authority to rubber-stamping. At the governance layer, the absence of pre-deployment review requirements and the diffusion of accountability undermine both IHL compliance and post-hoc redress. To address these failures we propose three mutually dependent controls: a contrastive explainability interface grounded in SHAP and Grad-CAM, a mandatory deliberation lock enforcing IHL-structured confirmation steps, and a Glass Box governance framework [2] enabling independent audit. Together, these restore the conditions for meaningful human control without requiring access to classified model internals.

1 Introduction

Habsora is an AI-targeting system used by the Israeli Defense Force (IDF), developed primarily by Unit 8200 within the Military Intelligence Directorate. The system is an ensemble of multiple specialised algorithms fused together [3] (Figure 5). First deployed during the May 2021 Operation ‘Guardian of the Walls’, it generates target recommendations using a multimodal intelligence pipeline that encompasses signal intelligence, telephone metadata, satellite and drone imagery, social graph analysis, and life-pattern modelling [3, 4]. During the 2023-2024 Gaza conflict, Habsora was reported to generate over one hundred target recommendations per day, which previously would require approximately twelve months of manual analysis [4, 5].

Habsora’s AI capability is the probabilistic classification and prioritisation of locations as military objectives. It assigns confidence scores derived from large-scale pattern matching across a shared intelligence pool, rather than positive identification in the traditional intelligence analysis sense [6]. Military commanders authorise each recommendation, yet the review window documented for each target was approximately only twenty seconds [4].

This creates a stakeholder dynamic with both ethical and legal significance. Its output influences its operators, who operate under high cognitive load and time pressure, the civilians in the target environment bearing the lethal consequences, and the broader international community, which depends on International Humanitarian Law (IHL) for the legitimacy of armed conflict. Steen et al. [7] offer a framework for this tension. They propose a two-axis model of Meaningful Human Control (MHC) which distinguishes between high automation and genuine human control. Both require adequate information and real freedom of action. The central argument of this report is that Habsora structurally undermines both conditions, and that this gap carries cascading implications across the technical, sociotechnical, and governance layers identified by Verdiesen et al. [1] in their Comprehensive Human Oversight Framework (CHOF).

2 Ethical Issue Identification

Verdiesen et al. [1]’s CHOF can be used to assess accountability for autonomous systems whose internals are not directly inspectable. It divides oversight into a 3×3 matrix: before, during and after deployment across a technical, socio-technical and governance mapping. This approach addresses gaps in accountability which would remain hidden if human oversight were treated as a single requirement. Applying this framework to Habsora’s implementation we get Figure 1.

	Before	During	After
Technical	Partial Built; no XAI embedded	Absent No reasoning chain shown	Absent No post-strike trace
Sociotechnical	Absent Civilian values not elicited (VSD)	Absent 20-sec rubber stamp dynamic	Absent Responsibility diffused across actors
Governance	Absent No EIA or legal pre-approval	Absent No IHL compliance check	Absent No audit or accountability

Figure 1: The CHOF framework for structuring ethical analysis across governance, socio-technical, and technical layers. The single yellow cell indicates where Habsora can be given partial compliance. The red cells indicate absence.

2.1 Technical Layer

Habsora is an ensemble of deep learning models processing multimodal data [3, 4]. Such architectures are inherently opaque; Arrieta et al. [8] characterise this as the accuracy/interpretability trade-off, where deep ensembles maximise predictive performance at the cost of transparency. While post-hoc explainability methods exist, reports on Habsora’s operational use indicate that operators were not required to examine the basis for the system’s recommendations, reducing human review to a cursory verification that a target was male [4].

This opacity directly links to legal consequences. Schmitt and Thurnher [6] establish that IHL’s distinction principle requires positive identification of a target as a military objective, alongside a proportionality assessment that is understood by the user. Without any interpretability, the legal review is effectively empty.

These failures map directly onto the highest-ranked principles identified by Khan et al. [9]. The opacity of Habsora’s recommendations and the absence of operator-facing explanations constitute a direct violation of **P-01** (transparency), while the inability to trace or attribute targeting decisions undermines **P-03** (accountability). The principle violations are further compounded by two structural challenges: the lack of technical understanding among decision-makers (**CH-06**), which prevents operators from critically evaluating system outputs, and the absence of binding legal frameworks

(CH-09), which removes any external obligation to address these gaps. Together, this creates a system that cannot provide the context necessary for meaningful human understanding or oversight.

2.2 Socio-technical

Value Sensitive Design (VSD) is a framework for incorporating human values into the design of technical systems in a principled and structured way [10, 11]. Central to VSD is the distinction between direct stakeholders, those who interact with a system, and indirect stakeholders, those affected by it without ever using it. In the case of Habsora, those indirect stakeholders are civilians in the target environment, whose values were not kept in mind during development. They remain non-consenting parties who face the consequences of the system's use without recourse, reflecting a large asymmetry between those operating the system and those subjected to it.

For IDF operators, the conditions necessary for genuine human oversight are structurally absent. Steen et al. [7] argue that meaningful human control requires both sufficient information and meaningful freedom of action. Habsora undermines both: operators receive recommendations without transparency into the underlying reasoning, and short decision windows leave almost no room for critical evaluation. Miller [12] shows that without understanding what went into a decision, the explaine cannot meaningfully interrogate it. Combined with the large volume of targets the system generates, what would normally require extended deliberation and proportionality assessments is reduced to a button press. Former IDF operators have described this as 'rubber-stamping' [13] (those with authority merely approving algorithmic recommendations rather than making them themselves) confirming the presence of automation bias, where operators over-rely on AI outputs under time pressure [14].

The consequence is that IDF operators formally bear moral and legal responsibility for strikes they cannot meaningfully evaluate, while the system's opacity diffuses accountability. This prevents the human control that International Humanitarian Law demands, and that Steen et al. [7] identify as a precondition for attributing responsibility.

2.3 Governance

Article 36 of Protocol I to the Geneva Conventions requires states to perform legal reviews of new weapons systems [15]. Israel is one of the few countries that have not ratified Protocol I [16], meaning they are not obligated to consider the legal consequences of systems like Habsora.

UNESCO's 2021 Recommendation on the Ethics of AI urges member states against delegating life-or-death decisions to automated systems without meaningful human oversight [17], but this instrument is non-binding. Israel's stated position that existing IHL is sufficient to govern autonomous weapon systems and that no new regulatory framework is required [18, 19] suggests that such recommendations carry little institutional weight in its operational decisions.

Once deployed, establishing accountability becomes difficult. Under IHL, operators of such systems are legally responsible for how weapons are used, even when autonomous functions are involved [20], and are judged on what they knew at the time [21]. In practice, responsibility is spread across developers, the algorithm, the receiving officer, and the approving commander [22]. Lavender's documented 10% error tolerance in targeting recommendations suggests Habsora may carry a similarly high error tolerance. Real-time monitoring during active conflict compounds the issue further.

After deployment, review processes are structurally undermined. The UN Commission of Inquiry on the Occupied Palestinian Territory has requested access to Gaza to investigate and preserve evidence, but Israel has refused to cooperate [23]. The opacity of Habsora further compounds this: without interpretable audit trails, it is not possible to establish in retrospect whether a casualty resulted from an AI recommendation or a human decision.

3 Responsible Solution Design

3.1 Technical

One technical failure identified in Section 2 is that operators receive target recommendations without a reasoning chain. The direct remedy would be a transparency-by-design philosophy, but given

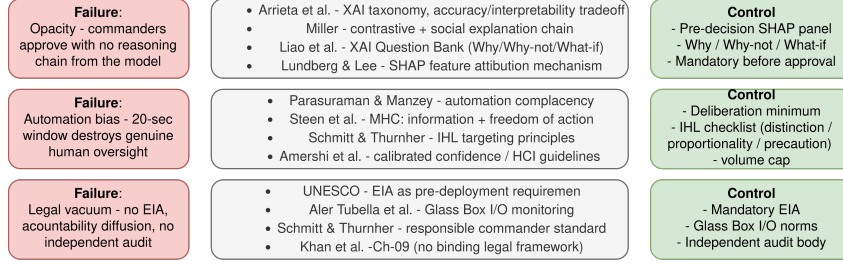


Figure 2: Framework-to-solution map: each identified failure mode (left) linked to the diagnosing it (centre) and the proposed control (right).

that these models would trigger the accuracy/interpretability trade-off [8], this adoption would be unlikely. A more realistic alternative would be implementing post-hoc explainability, integrated into the operators user interface. With this, as noted by Arrieta et al. [8]’s audience-centred definition: explainability is always relative to a given audience, the implementation should be calibrated to the field-operator, not for a Machine Learning engineer debugging the model weights.

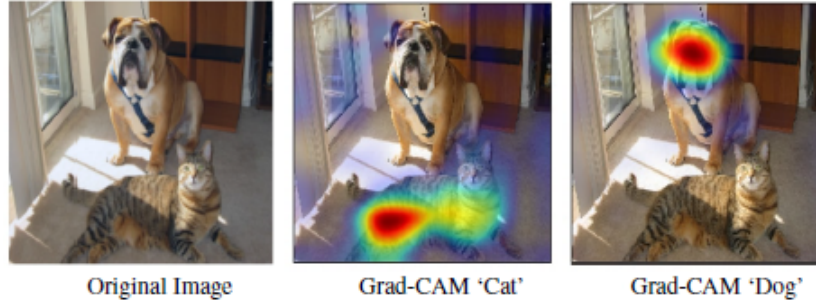


Figure 3: Canonical Grad-CAM example [24]. Class heatmaps localise image regions for ‘cat’ and ‘dog’. Applied to ‘Habsora’, this method would highlight satellite or drone imagery for classification.

We propose two post-hoc interpretability methods, SHapley Additive exPlanations (SHAP) [25] and Gradient-weighted Class Activation Mapping (Grad-CAM Figure 3)[24]. SHAP assigns each input feature a contribution score for a prediction. For example, a targeting recommendation for a building can be driven for 47% by the targets movement patterns, for 30% by their phone’s metadata, and for 23% by using drone footage showing targets entering it. Being model-agnostic (such as KernelSHAP), it can be applied to Habsora’s ensemble without accessing the models internals.

However, raw feature attribution is necessary but not sufficient. Miller [12] identifies four properties of human explanation: they’re contrastive (why X rather than Y), selective (not exhaustive), causal rather than probabilistic, and involve a social process between the explainer and explainee. SHAP’s raw output does not satisfy any of these properties and must therefore be expanded upon by providing the operator with a contrastive explanation: not only recommending the ‘best’ target, but also presenting the nearest non-target alternative alongside its scores. This allows the operator to evaluate ‘why this target rather than no strike’, critically assess the recommendation, and override it if necessary.

Liao et al. [26]’s XAI ‘QuestionBank’ operationalises this further. From the ten categories in their taxonomy, three are decision-critical for a strike: ‘why’, ‘why not’ and ‘what if’. These align with Miller [12]’s contrastive account (why X rather than Y) and map directly onto IHL’s targeting principles: *Why* supports distinction, *Why not* supports proportionality, and *What if* enables the precautionary assessment.

3.2 Socio-technical

As noted earlier, even if the technical layer would provide perfect explanations, the operational tempo would still undermine meaningful use. Steen et al. [7] define two conditions of moral responsibility:

adequate information and meaningful freedom of action. Our proposed mechanism is a mandatory minimum deliberation time, with a hard interface lock. The operator cannot approve any strike before the review period has elapsed, ensuring that the deliberation standard is met by design, and preventing premature confirmation.

The content of this deliberation follows IHL’s targeting principles. Rather than a single button press, the workflow should decompose into separate confirmations: *distinction* (is this a valid military target?), *proportionality* (is the anticipated military advantage proportionate to the expected civilian harm?), and *precaution* (have all measures to minimise civilian harm been made?). Requiring a confirmation per principle implements the legal standard identified by Schmitt and Thurnher [6], and prevents a multi-step judgement into a single button press.

Moreover, Parasuraman and Manzey [14] demonstrate that automation complacency is most pronounced when automation is highly and consistently reliable and when operators face competing task demands. The interface must actively interrupt this cycle by requiring engagement checks in the explanation layer: an operator should not be able to click through the IHL checklist without first reviewing the contrastive explanations and feature weights. This creates a forced interaction with the XAI output, preventing the explanation layer from becoming an obstacle to be bypassed rather than a genuine deliberative step.

Amershi et al. [27]’s human-AI interaction guidelines further require that the system communicate how well it can perform, calibrating displayed confidence to actual reliability and surfacing uncertainty explicitly. This directly counters automation bias by ensuring that operator trust is proportionate to system performance rather than assumed.

Lastly, Friedman et al. [11]’s VSD methodology requires that indirect stakeholders’ values are embedded in the system’s design. Civilian presence in the target environment cannot remain pure metadata: value-laden indicators such as civilian density or protected-site proximity should be surfaced in the operator interface as a structural requirement.

3.3 Governance

Governance interventions must be staged across the whole lifecycle of the systems, as reflected in the temporal axis of the CHOF [1] and the staged ethics-assessment approach advocated by Wasilow and Thorpe [22].

3.3.1 Pre-deployment

A mandatory pre-deployment ethical impact assessment (EIA) should be grounded in UNESCO’s recommendation against delegating life-or-death decisions to AI [17]. This ensures that human rights implications are systematically reviewed before deployment. Binding policy-level requirements are necessary to prevent the kind of gaps in governance identified in Khan et al. [9]’s **P-03** accountability and **P-05** autonomy, regarding the absence of accountability mechanisms and the erosion of human agency in AI systems.

3.3.2 During deployment

Glass Box compliance norms [2] should be followed during deployment: documented input/output thresholds for the system’s recommendations, mandatory escalation criteria for when human operators must intervene, and post-strike logs recording which recommendation was authorised and on what basis. Verdiesen et al. [1] integrate this approach into the governance layer of the CHOF: we argue it should not remain an optional design choice, but rather an operational legal requirement.

3.3.3 Post-deployment

An independent auditor should have access to the Glass Box logs to verify compliance with International Humanitarian Law and identify any violations. This would address the responsibility diffusion problem by replacing it with a traceable chain of authorisation, consistent with the responsible-operator standard set out by Schmitt and Thurnher [6]. This would restore accountability and the possibility of redress for affected parties, addressing the gaps identified in Khan et al. [9]’s **P-01** transparency and **P-03** accountability: the audit trail makes the authorisation chain inspectable and traceable.

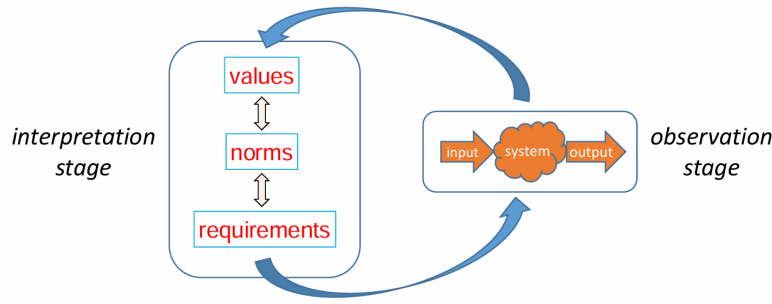


Figure 4: The Glass Box approach translates abstract values, such as human dignity, into observable norms and requirements to be implemented in the the AI system [1, 2]

4 Discussion and Conclusions

The three proposed controls (a pre-decision explainability layer, mandatory deliberation with IHL checks, and Glass Box governance) are mutually dependent and only function as a system. SHAP and Grad-CAM outputs are meaningless if operators are not given sufficient time to engage with them; deliberation time is insufficient if the information presented during it is opaque; and neither control is verifiable without the audit trail that Glass Box governance provides. This interdependency is consistent with Morley et al. [28]’s finding that post-hoc explainability is typically only implemented at the testing stage, leaving deployment and monitoring underdeveloped. Addressing Habsora’s failures therefore requires all three controls to be implemented together, each mirroring a distinct layer of the CHOF as proposed by Verdiesen et al. [1].

4.1 Stakeholder impact analysis

For IDF operators, contrastive XAI explanations and mandatory deliberation locks would restore genuine freedom of action: they would receive the information necessary to critically evaluate each recommendation, and the interface lock would guarantee sufficient time to do so. Should a strike ever be investigated by an independent auditor or international tribunal, operators could demonstrate precisely how they arrived at their decision, rather than having no legal defence. For civilians, mandating civilian density and protected-site proximity as structural interface requirements would embed their interests directly into the decision workflow. Combined with IHL checklist completion and appropriate handling of low-confidence recommendations, this could meaningfully reduce civilian casualties, while the audit trail would enable accountability for unjustified harm. For the international community, the Glass Box framework would enable independent verification of a classified system’s compliance record within an IHL-accountability context, shifting responsibility diffusion towards a structured and auditable chain of authorisation.

4.2 Concluding remarks and future perspectives

These recommendations carry limitations. The Glass Box approach depends on political will to establish an independent audit body, something Israel’s non-ratification of Additional Protocol I suggests may be difficult to achieve. Deliberation standards and volume constraints would also reduce operational tempo, a primary advantage motivating AWS adoption. There is a clear need for international binding frameworks and global standardisation of pre-deployment review requirements for AWS. Habsora is among the most extensively documented deployments of an AI targeting system under active IHL scrutiny, and the governance gaps it exposes are not unique to Israel. Without binding international frameworks establishing pre-deployment review requirements for autonomous weapon systems, the structural failures identified here will recur in every future deployment that prioritises operational tempo over meaningful human control.

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Appendix

Figures

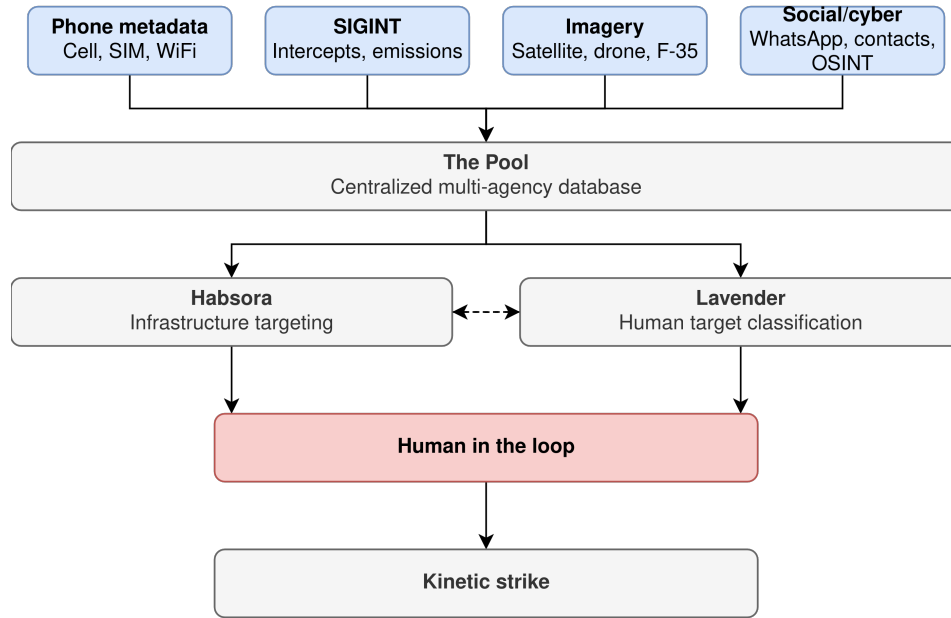


Figure 5: Habsora’s end-to-end pipeline, from multi-source intelligence ingestion through to strike, with the human-in-the-loop review stage highlighted.

Glossary

Term	Definition
Automation bias	The tendency of human operators to over-rely on automated system outputs, even when they have the authority to override them [14].
Contrastive explanation	An explanation that answers “why X rather than Y” by comparing a decision against an alternative outcome [12].
Ensemble architecture	A machine learning system composed of multiple specialised models whose outputs are combined to produce a unified prediction.
Glass Box	A compliance monitoring approach that defines observable norms at a system’s input/output boundary, enabling oversight without access to internal model logic [1].
Indirect stakeholder	A party affected by a system’s outputs without directly interacting with it [11].
Positive identification	The confirmed determination that a target is a legitimate military objective, as required by IHL’s distinction principle [6].
Post-hoc explainability	Explanation methods applied after a model has made a prediction, rather than built into the model’s architecture [8].

Abbreviations

AI	Artificial Intelligence
AWS	Autonomous Weapon Systems
CH-06	Challenge 06: Lack of technical understanding among decision-makers [9]
CH-09	Challenge 09: Absence of binding legal frameworks [9]
CHOF	Comprehensive Human Oversight Framework [1]
EIA	Ethical Impact Assessment
Grad-CAM	Gradient-weighted Class Activation Mapping
HUD	Head-Up Display
IDF	Israel Defense Forces
IHL	International Humanitarian Law
MHC	Meaningful Human Control [7]
P-01	Principle 01: Transparency [9]
P-03	Principle 03: Accountability [9]
SHAP	SHapley Additive exPlanations [25]
VSD	Value Sensitive Design [10, 11]
XAI	Explainable Artificial Intelligence